

EXP-10

CODE:

```
clc;

clear;

close all;

%% PN Sequence Lengths
PN_length = [3 9 27 81 243 729 2187];

%% Processing Gain (Linear)
PG_linear = PN_length;

%% Processing Gain in dB
PG_dB = 10 * log10(PG_linear);

%% Display Table
T = table(PN_length', PG_linear', PG_dB', ...
    'VariableNames', {'PN_Length','Processing_Gain_Linear','Processing_Gain_dB'});

disp('Processing Gain Table');

disp(T);

%% Plot

figure;

stem(PN_length, PG_dB, 'filled');

grid on;

xlabel('PN Sequence Length');

ylabel('Processing Gain (dB)');

title('Spread Spectrum: Processing Gain vs PN Sequence Length');

%% Display Results

fprintf('\nPN Sequence Length and Processing Gain:\n');
```

```

for i = 1:length(PN_length)
    fprintf('PN Length = %d, Processing Gain = %.2f dB\n', ...
        PN_length(i), PG_dB(i));
end

```

OUTPUT:

